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The stationary part which has armature windings mounted in it is called stator. It has three components, namely stator frame, stator core and armature winding.

Stator frame

This is the outer frame used for holding stator core and armature windings in proper position. Stator frame provides best ventilation with the help of holes provided in the frame itself (Figure 4.28).

Stator core

Stator core or armature core is made up of iron or steel alloy. It is a hollow cylinder and is laminated to minimize eddy current loss. The slots are cut on inner surface of the core to accommodate armature windings.

Armature winding

Armature winding is the coil, wound on slots provided in the armature core. One or more than one coil may be employed, depending on the type of alternator.

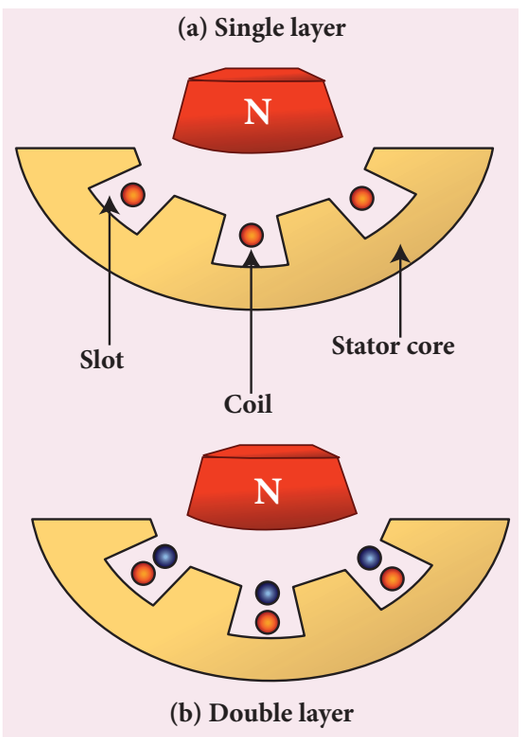


Figure 4.29 Armature windings

Two types of windings are commonly used. They are i) single-layer winding and ii) double-layer winding. In single-layer winding, a slot is occupied by a coil as a single layer. But in double-layer winding, the coils are split into two layers such as top and bottom layers (Figure 4.29).

ii) Rotor

Rotor contains magnetic field windings. The magnetic poles are magnetized by DC source. The ends of field windings are connected to a pair of slip rings, attached to a common shaft about which rotor rotates. Slip rings rotate along with rotor. To maintain connection between the DC source and field windings, two brushes are used which continuously slide over the slip rings.

There are 2 types of rotors used in alternators i) salient pole rotor and ii) cylindrical pole rotor.

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Salient pole rotor

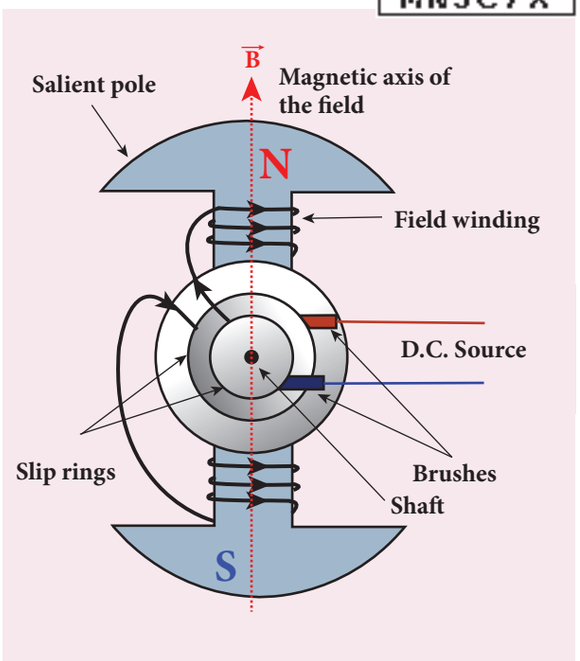


Figure 4.30 Salient 2-pole rotor

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